Sumset Growth, Boolean Coordinates, and Polynomial Counting

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A finite set can carry two very different kinds of algebra. If its elements are integers, adding two of them produces a sumset whose size reflects additive structure. If its elements are subsets of a fixed universe, taking symmetric difference produces a vector space over $\mathbb{F}_2$. Both settings transform set manipulation into algebra, but they make distinct predictions about how quickly a set grows under repeated operations.[1]

§1.1 A sharp extremal bound from sums and differences

Let $A = \{a_1 < \cdots < a_k\} \subseteq \{1, \ldots, 2021\}$, and define

$$S = A + A = \{a + b : a, b \in A\}, \qquad T = A - A = \{a - b : a, b \in A,\ a \geq b\}.$$

Suppose $S \cap T = \varnothing$. The $2k - 1$ sums

$$2a_1 < a_1 + a_2 < 2a_2 < \cdots < a_{k-1} + a_k < 2a_k$$

are distinct. Thus $|S| \geq 2k - 1$. The k nonnegative differences

$$0 < a_2 - a_1 < \cdots < a_k - a_1$$

are also distinct, giving $|T| \geq k$. As the two sets are disjoint and lie in $\{0, \ldots, 2a_k\}$,

$$3k - 1 \leq |S \cup T| \leq 2a_k + 1 \leq 4043.$$

Therefore $k \leq 1348$.

The bound is attained:

$$A = \{674, 675, \ldots, 2021\}.$$

Its nonnegative difference set is $\{0, \ldots, 1347\}$, while its sumset is $\{1348, \ldots, 4042\}$. They are disjoint, and $|A| = 1348$.

The calculation is instructive because every inequality is sharp at the same configuration. The extremal set is a long arithmetic progression. Its sumset is as small as an integer sumset can be, and its large starting point separates the sums from the differences.

§1.2 Why arithmetic progressions have minimal sumset growth

For any two finite nonempty subsets $A, B \subset \mathbb{Z}$, write $A = \{a_1 < \cdots < a_r\}$ and $B = \{b_1 < \cdots < b_s\}$. The chain

$$a_1 + b_1 < a_2 + b_1 < \cdots < a_r + b_1 < a_r + b_2 < \cdots < a_r + b_s$$

[1] For additive-combinatorial bounds and their polynomial proofs, see Tao and Vu, *Additive Combinatorics*, and Alon, *Combinatorial Nullstellensatz*, https://www.math.tau.ac.il/~nogaa/PDFS/comb_null.pdf.

contains $r + s - 1$ distinct elements of $A + B$, so

$$\boxed{|A + B| \geq |A| + |B| - 1.}$$

For arithmetic progressions with the same positive common difference, equality holds.

The inequality has a geometric interpretation: adding a second independent choice usually expands a set. When equality holds, the choices are aligned so that many distinct pairs yield the same sum. More elaborate inverse theorems in additive combinatorics ask whether small $|A + A|$ forces a set to resemble a generalized arithmetic progression. The extremal example above is the simplest visible case of that question.

§1.3 The finite-field version is Cauchy–Davenport

Over $\mathbb{F}_p$, where p is prime, addition wraps around. The correct bound for nonempty $A, B \subseteq \mathbb{F}_p$ is

$$\boxed{|A + B| \geq \min(p, |A| + |B| - 1).}$$

This is the Cauchy–Davenport theorem. The appearance of p reflects the impossibility of a sumset containing more than the entire field.

There is a striking polynomial proof of the essential case $|A|+|B|-2 < p$. Put $r = |A|$, $s = |B|$, and suppose for contradiction that $|A+B| \leq r+s-2$. Choose a set $C \supseteq A+B$ of exactly $r + s - 2$ field elements, and form

$$P(X, Y) = \prod_{c \in C} (X + Y - c).$$

The polynomial vanishes on every point of $A \times B$. Yet the coefficient of $X^{r-1}Y^{s-1}$ is

$$[X^{r-1}Y^{s-1}]P = \binom{r + s - 2}{r - 1},$$

which is nonzero modulo p, since $r + s - 2 < p$.

The contradiction comes from polynomial interpolation. For any polynomial P of total degree at most $r + s - 2$, the coefficient above equals

$$\sum_{\substack{a \in A \\ b \in B}} \frac{P(a, b)}{\prod\limits_{a' \in A \setminus \{a\}} (a - a') \prod\limits_{b' \in B \setminus \{b\}} (b - b')}.$$

To see this, apply the one-variable Lagrange interpolation formula first in X and then in Y; terms with total degree below $r + s - 2$ cannot contribute to the requested bidegree. Every summand on the right is zero because P vanishes on the grid, contradicting the nonzero binomial coefficient.

If $r+s-1 \geq p$, choose smaller nonempty subsets $A' \subseteq A$, $B' \subseteq B$ with $|A'|+|B'|-1 = p$, and apply the preceding argument to show $A' + B' = \mathbb{F}_p$. This establishes the general bound.

An additive-combinatorial inequality has become a statement about the coefficient of a polynomial. The fields $\mathbb{Z}$ and $\mathbb{F}_p$ behave differently, but the same cardinality growth has an algebraic explanation in both.

§1.4 Subsets of a universe are vectors over $\mathbb{F}_2$

Now let $U = \{1, \dots, n\}$, and associate to each $A \subseteq U$ its indicator vector

$$\mathbf{1}_A = (\mathbf{1}_{1\in A}, \dots, \mathbf{1}_{n\in A}) \in \mathbb{F}_2^n.$$

Under this encoding, symmetric difference becomes vector addition:

$$\mathbf{1}_{A\triangle B} = \mathbf{1}_A + \mathbf{1}_B \pmod 2.$$

Indeed an element belongs to $A\triangle B$ precisely when it belongs to exactly one of A, B.

The collection $\mathcal{P}(U)$ is therefore an n-dimensional vector space over $\mathbb{F}_2$. It has 2^n elements, and an independent family of r subsets generates exactly 2^r distinct symmetric differences.

For example, let $U = \{1, 2, 3\}$ and take $A = \{1, 2\}$, $B = \{2, 3\}$. Then

$$A\triangle B = \{1, 3\}, \qquad \langle A, B\rangle_\triangle = \{\varnothing, A, B, \{1, 3\}\}.$$

Only four distinct subsets appear because the two indicator vectors generate a two-dimensional subspace.

Unlike ordinary sumsets in the integers, a vector subspace can be perfectly stable under addition: $V + V = V$. A small doubling set may therefore signal algebraic closure rather than the geometry of a progression, depending on the ambient group.

§1.5 Venn diagrams are coordinate decompositions

Given m subsets $A_1, \dots, A_m$ of U, each element $u \in U$ has a membership signature

$$\sigma(u) = (\mathbf{1}_{u\in A_1}, \dots, \mathbf{1}_{u\in A_m}) \in \{0, 1\}^m.$$

The 2^m possible signatures partition U into disjoint Boolean atoms. Every expression built from unions, intersections and complements is a union of some of these atoms.

This gives a rigorous algebraic alternative to visually reading a complicated Venn diagram. For example,

$$A \setminus (B \cup C) = A \cap B^c \cap C^c$$

selects exactly the signature $(1, 0, 0)$. The proof is a one-line statement about membership coordinates.

There is also a useful distinction between two operations. Intersection acts like coordinatewise multiplication of indicator vectors, while symmetric difference acts like addition modulo two. These are precisely the ring operations of a Boolean ring. Its algebra is different from integer addition, but both arise from asking which combinations of finite objects preserve information and which combinations force growth.